

# Student Supervision & Teaching

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## THESIS AND INTERN SUPERVISION:

Löwe, H. (2023). *Implementation of correlation between PFS and OS into the **simIDM** package*. Three month full-time internship with Roche Statistical engineering, Basel. Co-supervised with Daniel Sabanés Bové and Matthias Meller.

Vazquez Rabunal, M. (2023). *Improved treatment effect estimation in subgroups displayed in forest plots based on Bayesian shrinkage priors*. Four month full-time internship with Roche Biostatistics, Basel. Co-supervised with Marcel Wolbers and Daniel Sabanés Bové.

Sawant, A. (2022). *Truncation by death for longitudinal continuous data: estimands and estimation methods*. Four month full-time internship with Roche Biostatistics, Basel.

Chen, X. (2019). *Sample size for prediction models*. Three month full-time internship with Roche Biostatistics, Basel. Co-supervised with Markus Elze and Gian Thanei.

Li, K. (2018). *Optimism-corrected treatment effect estimates in subgroups displayed in forest plots for time-to-event outcomes*. Master thesis to obtain Master in Biostatistics, University of Zurich. Initiated within three month full-time internship with Roche Biostatistics, Basel. Co-supervised with Marcel Wolbers. Academic supervisor: Prof. Torsten Hothorn, University of Zurich.

Meller, M. (2016). *Joint Modelling of Overall Survival and Progression Free Survival*. Master thesis to obtain Master in Biostatistics, University of Ulm. Initiated within four month full-time internship with Roche Biostatistics, Basel. Academic supervisor: Prof. Jan Beyersmann, University of Ulm. Winner of the Bernd Streitberg Award 2018 of the German Region of the International Biometric Society.

Hebeisen, M. (2014). *Estimation from Interval-censored Time-to-event Data: Method Comparison by Simulation based on GALLIUM Study for Follicular Lymphoma*. Master thesis to obtain Master in Biostatistics, University of Zurich. Prepared within seven month full-time internship with Roche Biostatistics, Basel. Academic supervisor: Prof. Torsten Hothorn, University of Zurich. download thesis

Hubeaux, S. (2013). *Analysis of time-to-event data with interval- or left-censored covariates*. Six month full-time internship with Roche Biostatistics, Basel.

Boutellier, P. (2013). *What does a PSA bounce look like? A numerical and graphical description of the prostate-specific antigen (PSA) bounce phenomenon in patients treated for prostate cancer with brachytherapy only*. Master thesis to obtain Master in Medicine, University of Zurich.

Loiero, D. (2012). *Inter- and Intrareader variability of Ratingen score in the Swiss Clinical Quality Management database*. Master thesis to obtain Master in Medicine, University of Zurich.

Scherer, A. (2011). *Determinants of TNFa inhibitor prescription in the Swiss Clinical Quality Management cohort of axial Spondylarthropathy patients*. Semester thesis in Master of Biostatistics program, University of Zurich.

Achermann, R. (2008). *Netrin-1 und DCC Expression bei Hirntumoren*. Semester thesis in Master of Biostatistics program, University of Zurich.

Lanz, K., Weber M. (2005). *Schleistung unter körperlicher Anstrengung*. Swiss Technical School for Optometrics, Olten, Switzerland.

## TEACHING LEADERSHIP

- At Roche, I led the official roll-out of the ICH E9 addendum within Roche, for statisticians and partner functions (such as clinical science, regulatory, biometrics, early research).
- A full-day introductory biostatistics course that I gave at Roche together with Marcel Wolbers has been selected as part of the official training program for all Roche affiliates globally.
- At Roche, I co-initiated a series of regular talks for statisticians and non-statisticians on a broad range of topics, reaching a broad audience across the company.
- At Merck, I initiated a series of regular talks for statisticians and non-statisticians on a broad range of topics, reaching a broad audience across the company.

## SHORT COURSES FOR STATISTICALLY TRAINED AUDIENCES

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| 06/2024 | <i>Introduction to and applications of survival analysis</i> . (IBS German and Austrian-Swiss Region Summer school, with Paul Blanche [U Copenhagen], Sarah Friedrich [U Augsburg], Helga Wagner [U Linz].)                                                                                               |
| 05/2024 | <i>Courses on basics of clinical trials, group-sequential designs, estimands, subgroup analysis</i> . (Within MSc program "Master of Pharmaceutical Sciences in Drug Development and Regulatory Affairs", University of Innsbruck, Austria.)                                                              |
| 09/2023 | <i>Advanced group-sequential and adaptive confirmatory clinical trial designs, with R practicals using rpact</i> . (Pre-conference course at 5th Conference of the Central European Network, Basel, Switzerland, with Gernot Wassmer [rpact], Marc Vandemeulebroecke [Novartis], Marcel Wolbers [Roche].) |
| 08/2023 | <i>Causal Inference in Randomized Controlled Trials</i> . (Pre-conference course at Joint Statistical Meetings in Toronto, with Mouna Akacha, Robin Dunn, Tianmeng Lyu [all Novartis], Shanti Gomati [FDA].)                                                                                              |
| 08/2023 | <i>Basic methods for group-sequential trials</i> . (Roche internal course for biostatisticians, Mississauga, Canada.)                                                                                                                                                                                     |
| 03/2023 | <i>Courses on basics of clinical trials, group-sequential designs, estimands, subgroup analysis</i> . (Within MSc program "Master of Pharmaceutical Sciences in Drug Development and Regulatory Affairs", University of Innsbruck, Austria.)                                                              |
| 02/2023 | <i>Some challenges with implementing estimands in real life</i> . (Recording for online training for <i>The Effective Statistician Academy</i> , Butzbach, Germany.)                                                                                                                                      |
| 02/2023 | <i>Basic and advanced methods for group-sequential trials</i> . (Roche internal course for biostatisticians. With Marcel Wolbers.)                                                                                                                                                                        |
| 09/2022 | <i>Advanced group-sequential and adaptive confirmatory clinical trial designs, with R practicals using rpact</i> . (Organized by the Basel Biometrics Section, Basel. With Gernot Wassmer [rpact], Marc Vandemeulebroecke [Novartis], Marcel Wolbers [Roche], full-day course)                            |
| 02/2022 | <i>"Confirmatory adaptive designs" for drug developers</i> . (Two half-days course for drug development functions at Roche, run 3 times with in total >650 participants)                                                                                                                                  |

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| 12/2021           | <i>Adaptive Design in Clinical Trials.</i> (Two half-days course in collaboration with DIA)                                                                                                                                                    |
| 05/2021           | <i>successR: comprehensive resource for training, methodology, and computational tools around success probabilities.</i> (Five 2.5-hour shortcourses for all Roche biostatisticians)                                                           |
| 05/2021           | <i>Stop the abuse: A plea for a more principled approach to the analysis of time-to-event endpoints with competing risks, with a focus on analysis of AEs.</i> (Two 2.5-hour shortcourses for all Roche biostatisticians, with Marcel Wolbers) |
| 01/2021           | <i>Answering Old Questions with New Tools: Application of the ICH E9 Addendum in Oncology.</i> (Two 2.5-hour shortcourses for all Roche biostatisticians, with Godwin Yung)                                                                    |
| 12/2020           | <i>Answering Old Questions with New Tools: Application of the ICH E9 Addendum in Oncology.</i> (Half-day shortcourse at 76th Annual Deming Conference on Applied Statistics, with Evgeny Degtyarev)                                            |
| 06/2020 – 12/2020 | <i>A Toolbox for Innovative Confirmatory Trial Designs. Four 2h modules on group-sequential, adaptive, and multiarm multistage designs.</i> (Virtual course for all Roche biostatisticians, with Marcel Wolbers)                               |
| 10/2019           | <i>Advanced methods for group-sequential and adaptive trials.</i> (Summer school organized on behalf of the German Region of the IBS, Lambrecht, Germany. With Marc Vandemeulebroecke, Gernot Wassmer, and Marcel Wolbers, 3-day course)       |
| 07/2019           | <i>Advanced methods for group-sequential trials.</i> (ISCB40, Leuven, Belgium. With Marcel Wolbers. Half-day pre-conference course)                                                                                                            |
| 03/2019           | <i>Group-sequential clinical trial designs, advanced topics.</i> (Roche, Basel. With Marcel Wolbers. Half-day course)                                                                                                                          |
| 03/2018           | <i>Group-sequential and adaptive clinical trial designs, with R practicals.</i> (Organized by the Basel Biometrics Section, Basel. With Daniel Sabanés Bové, Marc Vandemeulebroecke, Marcel Wolbers, full-day course)                          |
| 06/2013           | <i>Analysis of Longitudinal data.</i> (Roche, Basel. 8-hour course)                                                                                                                                                                            |
| 01/2011           | <i>Analysis of Longitudinal data.</i> (Institut für Medizinische Biometrie, Epidemiologie, und Informatik, University of Mainz. 3-day course, with Leo Held)                                                                                   |
| 06/2010           | <i>Analysis of Longitudinal data, as part of Master in Medical Biometry.</i> (Department of Medical Biometry, University of Heidelberg, Heidelberg. 3-day course, with Leo Held)                                                               |

## GRADUATE LEVEL TEACHING FOR STATISTICIANS

Preparing and teaching lecture to Master students, preparing and discussing exercises (blackboard and with R), preparing and grading exams.

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|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12/2019           | <i>Introduction to Applied Mathematics and Informatics in Drug Discovery (AMIDD), one session: Interim look into pivotal clinical trials - why it matters and how to do it</i> (Graduate level, University of Basel) |
| 11/2013           | <i>Statistical Methods for Clinical Research, one session on group-sequential designs in clinical research</i> (Graduate level, University of Zurich)                                                                |
| 11/2012 – 12/2012 | <i>Statistical Methods for Clinical Research, two sessions: group-sequential designs in clinical research and measures of agreement</i> (Graduate level, University of Zurich)                                       |

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|-------------------|---------------------------------------------------------------------------------------------------------------------------|
| 09/2011 – 12/2011 | <i>Statistical Methods for Clinical Research</i><br>(Graduate level, University of Zurich)                                |
| 02/2010 – 06/2010 | <i>Longitudinal data analysis</i><br>(joint with Prof. L. Held, graduate level, University of Zurich)                     |
| 09/2009 – 12/2009 | <i>Biostatistical methods, Part II: Statistical methods for clinical trials</i><br>(graduate level, University of Zurich) |
| 02/2009 – 05/2009 | <i>Likelihood inference</i><br>(joint with Prof. L. Held, graduate level, University of Zurich)                           |
| 10/2005 – 02/2006 | <i>Introduction to Statistics</i> (undergraduate level, University of Bern)                                               |

#### SHORT COURSES FOR NON-STATISTICAL AUDIENCES

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|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09/2024           | <i>Survival analysis, group-sequential trial designs</i> (European Center of Pharmaceutical Medicine Course, Basel)                                                                                                |
| 06/2024           | <i>Basics of clinical trial designs</i> (European Center of Pharmaceutical Medicine Course, Basel)                                                                                                                 |
| 10/2023           | <i>Futility analyses - they are not futile!</i> (90min session for medical affairs, Roche, Basel)                                                                                                                  |
| 10/2023           | <i>Introduction to survival analysis</i> (30min session for medical affairs, Roche, Basel)                                                                                                                         |
| 05/2023           | <i>Multiplicity</i> (2h course for Pharmacometrics department, Roche, Basel)                                                                                                                                       |
| 09/2022 – 10/2022 | <i>Introduction to statistics, survival analysis, interpretation of subgroups in clinical trials, interim analyses</i> (2 days for swiss health quality association, Zug)                                          |
| 09/2022           | <i>The Concept of Estimands / Group-Sequential Designs and Interim Analyses in Drug Development / Survival Analysis</i> (European Center of Pharmaceutical Medicine Course, Basel)                                 |
| 06/2022           | <i>Basic Statistical Concepts</i> (European Center of Pharmaceutical Medicine Course, Basel)                                                                                                                       |
| 06/2022           | <i>Estimands in Multiple Sclerosis</i> (1-hour course for Roche's MS clinical science community, virtual event)                                                                                                    |
| 04/2022           | <i>Platform trials in drug development</i> (1-hour course for Roche's Nordic commercial organization, virtual event)                                                                                               |
| 11/2021           | <i>The final word on interim analyses</i> (2-hour courses for all product development colleagues, Roche, Basel, virtual event)                                                                                     |
| 10/2021           | <i>How to survive survival analysis?</i> (2-hour courses for all product development colleagues, Roche, Basel, virtual event)                                                                                      |
| 05/2021           | <i>Introduction to statistics.</i> (2-hour courses for all product development colleagues, Roche, Basel, virtual event)                                                                                            |
| 05/2021           | <i>2-hour training for clinical reviewers: Implications of the ICH E9 estimand addendum on how we develop, run, and analyse clinical trials</i> (Agência Nacional de Vigilância Sanitária (ANVISA, virtual event)) |
| 05/2021           | <i>Introduction to statistics.</i> (2-hour courses for all product development colleagues, Roche, Basel, virtual event)                                                                                            |
| 01/2021           | <i>Introduction to statistics.</i> (Two 2-hour courses for all Medical Science Liaison colleagues within Roche Medical Affairs, Roche, Basel, virtual event)                                                       |
| 12/2020           | <i>Survival analysis in rheumatology.</i> (1-hour course for medical doctors, organized by Roche Switzerland at St. Gallen Hospital, virtual event)                                                                |

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| 11/2020           | <i>Introduction to statistics.</i> (Two 2-hour courses for all product development colleagues, Roche, Basel, virtual event)                                                                                   |
| 09/2020           | <i>Introduction to survival and interim analyses.</i> (Half-day course for Regulatory Documentation Department, Roche, Basel & Welwyn, virtual event)                                                         |
| 08/2020           | <i>Introduction to statistics.</i> (Half-day course for Regulatory Documentation Department, Roche, Basel & Welwyn, virtual event)                                                                            |
| 01/2020           | <i>Introduction to survival and interim analyses.</i> (Half-day course for Regulatory Documentation Department, Roche, Basel)                                                                                 |
| 01/2020           | <i>How to interpret subgroup analyses in clinical trials?</i> (2-hour course for medical doctors, organized by Roche Switzerland at University Hospital Zurich.)                                              |
| 11/2019 – 01/2020 | <i>Design of Experiments &amp; Effective Statistical Practices.</i> (Three two-hour courses for early development organization, Roche, Basel)                                                                 |
| 10/2019           | <i>Introduction to statistics.</i> (Half-day course for Regulatory Documentation Department, Roche, Basel)                                                                                                    |
| 01/2019           | <i>Introduction to statistics in clinical trials.</i> (2-hour course for Medical Science Liaison group, Roche Swiss Affiliate, Reinach)                                                                       |
| 01/2019           | <i>Introduction to survival analysis and clinical trial designs, Part II.</i> (2-hour course for medical doctors, organized by Roche Switzerland at Baden Hospital.)                                          |
| 01/2019           | <i>Estimands and real-world data in drug development.</i> (Half-day course for early development Neuroscience, Ophthalmology and Rare Disease scientists, Roche, Basel)                                       |
| 11/2018           | <i>Introduction to survival analysis and clinical trial designs.</i> (Half-day course for early development Neuroscience, Ophthalmology and Rare Disease scientists, Roche, Basel)                            |
| 03/2018           | <i>Introduction to survival analysis and clinical trial designs.</i> (3-hour course for medical doctors, organized by Roche Switzerland at Baden Hospital.)                                                   |
| 02/2018           | <i>Introduction to survival analysis and clinical trial designs.</i> (Half-day course for Regulatory Documentation Department and Global Project Managers, Roche, Basel)                                      |
| 11/2017           | <i>Introduction to statistics.</i> (Half-day course for Regulatory Documentation Department and Global Project Managers, Roche, Basel)                                                                        |
| 07/2017           | <i>Sample size planning and survival analysis in clinical trials.</i> (Full-day course for Medical Affairs and Clinical Science Departments, Roche, Basel, jointly with Marcel Wolbers)                       |
| 04/2017           | <i>Introduction to statistics.</i> (Half-day course for Competitive Intelligence Group, Roche, Basel)                                                                                                         |
| 03/2017           | <i>Introduction to statistics.</i> (2-hour course for Finance Department, Roche, Basel)                                                                                                                       |
| 11/2016           | <i>Introduction to statistics.</i> (Half-day course for Regulatory Documentation Department, Roche, Mississauga)                                                                                              |
| 10/2016           | <i>Introduction to statistics.</i> (Half-day course for Regulatory Documentation Department, Roche, Basel)                                                                                                    |
| 08/2016           | <i>Introduction to statistics.</i> (3-hour course for Media Group Roche, Basel)                                                                                                                               |
| 03/2016 – 11/2016 | <i>Five workshops on specific biostatistical topics in clinical trials.</i> (3-hour courses for medical doctors, organized by Roche Switzerland at hospitals in Sion, St. Gallen, Frauenfeld, Aarau, Zürich.) |
| 05/2016           | <i>Introduction to statistics.</i> (Half-day course for Regulatory Documentation Department, Genentech, South San Francisco and Roche, Basel)                                                                 |

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| 02/2016           | <i>Introduction to statistics in clinical trials.</i> (Half-day course for Medical Science Liaison group, Roche Swiss Affiliate, Reinach)                                                                                                                                                             |
| 01/2016 – 02/2016 | <i>Introduction to statistics. Certificate of Advanced Studies (CAS) course, as part of the Master of Advanced Studies in Managed Health Care.</i><br>(2-day course, ZHAW Zurich University of Applied Sciences, School of Management and Law, Winterthur Institute for Health Economics, Winterthur) |
| 11/2015           | <i>Introduction to statistics.</i> (1-day course for Statistical Programming and Analysis Department, Roche Basel)                                                                                                                                                                                    |
| 10/2015           | <i>Introduction to statistics.</i> (Two 1-day courses for Regulatory Documentation Department, Roche Basel & Welwyn)                                                                                                                                                                                  |
| 05/2015           | <i>Introduction to statistics.</i> (1-day course for Product Development Department, Roche Basel)                                                                                                                                                                                                     |
| 02/2015           | <i>Introduction to statistics. Certificate of Advanced Studies (CAS) course, as part of the Master of Advanced Studies in Managed Health Care.</i><br>(2-day course, ZHAW Zurich University of Applied Sciences, School of Management and Law, Winterthur Institute for Health Economics, Winterthur) |
| 06/2014 – 11/2014 | <i>Four workshops on specific biostatistical topics in clinical trials.</i> (3-hour courses for medical doctors, organized by Roche Switzerland at hospitals in Zürich, Lausanne, Basel, Baden.)                                                                                                      |
| 01/2014 – 03/2014 | <i>Introduction to statistics. Certificate of Advanced Studies (CAS) course, as part of the Master of Advanced Studies in Managed Health Care.</i><br>(3-day course, ZHAW Zurich University of Applied Sciences, School of Management and Law, Winterthur Institute for Health Economics, Winterthur) |
| 11/2013           | <i>Workshop on survival analysis in oncology.</i> (3-hour course for Roche Oncology sales force, Reinach)                                                                                                                                                                                             |
| 06/2013 – 04/2014 | <i>Seven workshops on survival analysis in oncology.</i> (3-hour courses for medical doctors organized jointly by Roche Switzerland and Novartis Switzerland at hospitals in Basel, Zürich, Bern, Luzern, Lausanne, Lugano, Olten.)                                                                   |
| 01/2012 – 03/2012 | <i>Introduction to statistics. Certificate of Advanced Studies (CAS) course, as part of the Master of Advanced Studies in Managed Health Care.</i><br>(3-day course, ZHAW Zurich University of Applied Sciences, School of Management and Law, Winterthur Institute for Health Economics, Winterthur) |
| 09/2011           | <i>Statistics in oncology.</i><br>(1-day course, cancer biology PhD program, University of Zurich)                                                                                                                                                                                                    |
| 04/2011           | <i>How to write a scientific paper – statistical aspects.</i><br>(4-hour course as part of medical curriculum, University Hospital, Zurich)                                                                                                                                                           |
| 02/2011           | <i>Introduction to biostatistics.</i><br>(3-hour course, continuing education for radiologists, University Hospital, Bern)                                                                                                                                                                            |
| 03/2010           | <i>Samples, confidence intervals, statistical tests, and p-values.</i><br>(1-hour talk, Medical radiology department, University Hospital Zurich, Zurich)                                                                                                                                             |
| 01/2010 – 05/2010 | <i>Introduction to statistics. Certificate of Advanced Studies (CAS) course, as part of the Master of Advanced Studies in Managed Health Care.</i><br>(3-day course, ZHAW Zurich University of Applied Sciences, School of Management and Law, Winterthur Institute for Health Economics, Winterthur) |
| 09/2010           | <i>Statistics in oncology.</i><br>(1-day course, cancer biology PhD program, University of Zurich)                                                                                                                                                                                                    |

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| 09/2009           | <i>Statistics in oncology.</i><br>(1-day course, cancer biology PhD program, University of Zurich)                |
| 09/2009           | <i>Assessment of agreement between continuous variables.</i> (1-hour talk, Waid Hospital, Zurich)                 |
| 02/2009           | <i>Introduction to biostatistics.</i> (1-hour talk, Triemli Hospital, Zurich)                                     |
| 01/2009           | <i>Introduction to biostatistics.</i><br>(1-day course, continuing studies for researchers, University of Zurich) |
| 09/2008           | <i>Statistics in oncology.</i><br>(1-day course, cancer biology PhD program, University of Zurich)                |
| 02/2008           | <i>Linear models.</i><br>(1-day course, continuing studies for researchers, University of Zurich)                 |
| 04/2005 – 05/2005 | <i>Introduction to R.</i> (2-day course, continuing studies, University of Bern)                                  |

#### FURTHER SHORT COURSES

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| 09/2011 | <i>Optimization.</i><br>(2-hour course, Energy trading academy, BKW FMB Energie AG) |
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